

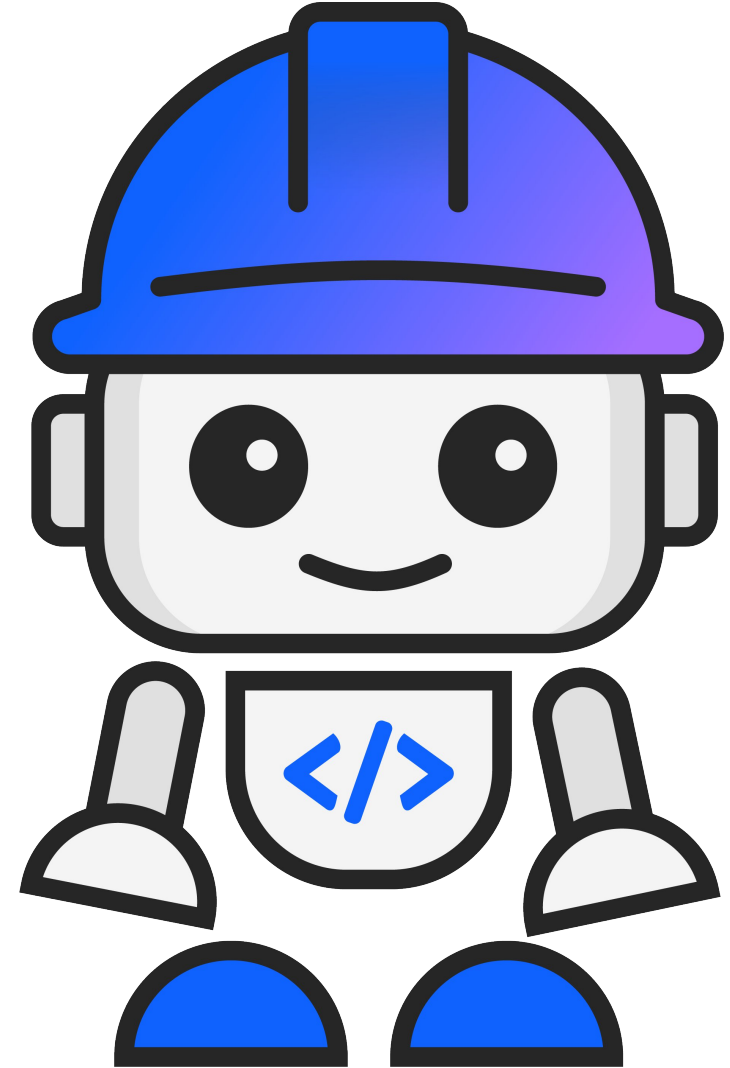
IBM Bob Premium Package for Java Modernization

*Streamline the Java application lifecycle, reduce cost
and risk in modernization*

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Introducing IBM Bob, your partner across the SDLC

Delivering AI across the SDLC requires a different architecture. Bob is built for that.



Hi, I'm Bob!

I can help you to

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The scale of the Java modernization problem

The problem isn't a few outdated applications

It's entire Java estates that have accumulated over decades

Legacy Java systems dominate enterprise environments and consume the majority of engineering effort

68% of IT budgets are spent maintaining legacy systems rather than building new capabilities

The Legacy Software Modernization **exceeded \$15 billion in 2025**, reflecting growing enterprise urgency

Engineering teams spend **60–75% of their time** maintaining existing systems, not delivering new value

70% of Fortune 500 software is **over 20 years old**, turning Java estates, much of it built on early Java architectures

Why modernization efforts stall

Teams know they need to modernize, but the complexity and risk prevent progress

Most teams don't lack motivation...

...they lack safe and structured ways to evolve complex systems

- Lack of visibility into dependencies, frameworks, and hidden coupling
- High refactor due to fragile configs, concurrency issues, and outdated libraries
- Poor documentation and limited regression coverage
- Tools surface static analysis noise, not architecture insight
- Fragmented tooling forces constant context switching across platforms
- Fear of unknown performance and operational impacts

The ideal modernization method is changing

Before: prior modernization

- Consulting-led programs
- Manual code analysis
- Fragmented toolchains
- Large, multi-year migration paths
- Modernization treated as a separate initiative

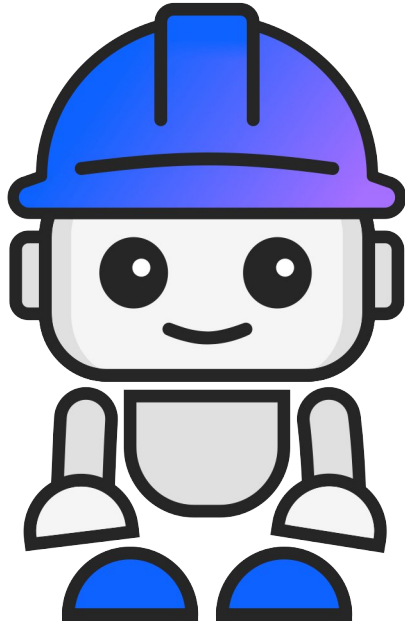
After: AI-assisted modernization

- ✓ **Developer-driven modernization**
- ✓ **AI-assisted reasoning across the codebase**
- ✓ **IDE-native workflows**
- ✓ **Incremental changes applied continuously**
- ✓ **Modernization becomes part of everyday development**

Modernization stops being a “project”
It becomes standard engineering work

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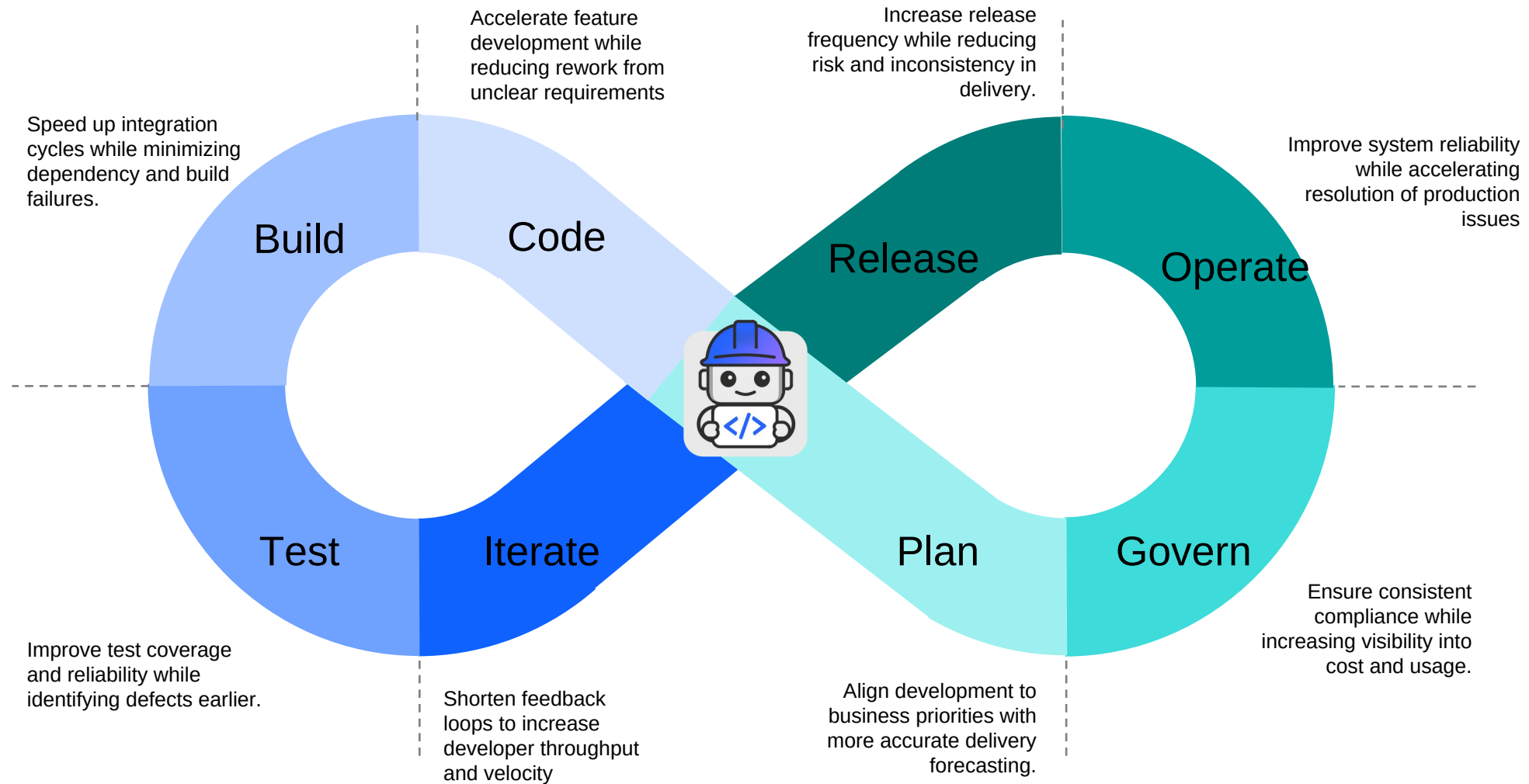


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What does Bob unlock for your business?

Translate coordinated delivery into measurable business value across the software lifecycle



IBM Bob Premium Package for Java Modernization

IBM Bob is a partner for enterprise developers that plans, executes, validates, and governs multi-step modernization tasks across the SDLC

The IBM Bob Premium Package for Java Modernization adds optimized workflows, improved security, and direct integrations.

Optimized workflows

Reduce token usage and increase quality of outputs common Java modernization workflows

Native integrations

Integrates AMA application insights into IBM Bob's end-to-end modernization workflow, turning Liberty migration into a governed, repeatable process.

Secure Modernization

Helps teams modernize applications with greater confidence by embedding security, validation, and risk management throughout the modernization process.

Three strategic workflows enabled with IBM Bob Premium Package for Java Modernization

Shift aging, business-critical Java applications from being a drag on innovation to becoming a catalyst for future growth

1. Java Upgrade

Workflow choreographs from Analysis through AI-Driven version upgrades and validations

Detailed summary and mermaid diagram turns complex, multi-step migrations into quick, scannable visuals

2. UI Modernization

Modernize legacy UI frameworks like JSF, Struts to sleek ReactJS library, instantly elevating the User Experience.

Handles the migration end-to-end, delivering a clean, production-ready codebase.

3. Liberty Replatforming

Replatform to Liberty Runtime using migration plan to guide the journey based on highly personalized analysis of specific estate.



Powered by Application Modernization Accelerator (AMA)** to streamline the WebSphere to Liberty migration, reducing cost & complexity



Liberty supports modern Java and Jakarta EE standards and frameworks, enabling teams to build cloud-native applications, with smaller memory footprint and stronger scalability.

***AMA is not included in the Bob Premium Package, it is licensed separately and only available with JSphere Suite: Enterprise Application Runtimes (EAR).*

IBM WebSphere Liberty is a modern, modular, and secure cloud-native Java runtime available in JSphere EAR.

How IBM Bob modernizes Java systems

Bob modernizes entire Java systems safely and incrementally

System-Level Understanding

Bob analyzes entire systems, not just individual files

Bob understands:

- Java runtimes
- Frameworks dependencies
- Application Servers
- Architectural layers
- UI and Integration Points

This enables modernization across **complete systems**, not isolated code fragments

Guided Developer Workflows

Modernization becomes a **structured engineering loop**:

- *Analyze*
- *Transform*
- *Build*
- *Test*
- *Validate*

Changes are applied in **small, verifiable increments**, reducing migration risk and enabling continuous progress

Enterprise Guardrails

Modernization must respect production constraints

Bob embeds governance through:

- Validation checkpoints
- Human approvals
- Traceable code changes
- Automated documentation

This ensures modernization remains **safe, auditable, and production-ready**

Modernizing Java should be engineering...not archaeology

Turn Java modernization into an ongoing governed pipeline

Understand the System

Analyze entire applications and environments

*Creates a **map of the system** before changes begin*



Transform with Confidence

Deterministic automation and guided upgrades

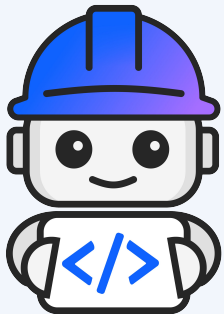
*Changes happen **incrementally and safely***



Govern the Change

Built-in validation ensures modernization remains production-ready

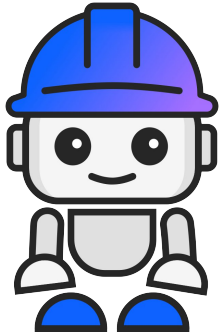
*Modernization stays **reliable, compliant, and auditable***



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Thank you



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